

Workshop Execution Plan: "Project: Engine Room/API Blueprint/any other name"

Target Audience: 1st-Year Students

Duration: 17 Days (15 Days Self-Learning + 2 Days Project Build)

Tech Stack: PERN (PostgreSQL, Express, React, Node.js) - *Backend-First Focus*

1. Executive Summary

This proposal outlines a 17-day, self-paced PERN stack workshop designed specifically for 1st-year students. This workshop adopts a strictly **Backend-First, Pure CRUD** approach. Students will spend the first 11 days mastering Node.js, Express, and PostgreSQL before being introduced to a pre-styled React frontend. The learning ecosystem will be driven entirely through a peer-to-peer WhatsApp community.

2. Core Strategy & Objectives

- **The Narrative:** The workshop will be branded as building the "Engine Room." We will cultivate a culture that backend development is elite, focusing on the mechanics of data flow rather than frontend aesthetics.
- **Peer-Driven Troubleshooting:** Mentors will step back and allow students to resolve each other's doubts to build a self-sustaining developer community.

3. Curriculum Blueprint (The 15-Day Sprint)

The learning materials will be highly curated, bite-sized, and divided into clear daily milestones.

Phase	Days	Focus Area	Key Deliverables / Milestones
Phase 1: Foundation	1 - 4	JS Basics, Async/Await, Node & Express Setup	Milestone 1: Running a local Express server that returns mock JSON data to Postman.
Phase 2: Database	5 - 9	PostgreSQL, raw SQL queries, pgAdmin	Milestone 2: Creating tables and executing raw SELECT and INSERT queries in pgAdmin.
Phase 3: Integration	10 - 11	pg pool, wiring Express routes to	Milestone 3 (Major): A fully functional backend

		SQL	CRUD API successfully tested via Postman.
Phase 4: The Frontend	12 - 15	React basics, useState, useEffect, fetch	Milestone 4: Wiring a basic UI to their custom backend API (handling CORS and state).

4. Operational Mechanics (WhatsApp Community)

To prevent the WhatsApp group from becoming chaotic or inactive, we will enforce strict community guidelines from Day 1.

The "SOS Protocol"

Students are not allowed to post generic complaints (e.g., "My code crashed"). All troubleshooting requests must include:

1. **The Screenshot:** A clear image of the error log and the specific code block.
2. **The Goal:** What they were trying to achieve.
3. **The Attempt:** What they have already done to try and fix it.

Mentor Guidelines: The "Wait & Validate" Strategy

- **The 45-Minute Rule:** Mentors are strictly forbidden from answering a student's doubt for the first 45 minutes. This creates a vacuum that encourages peers to step up.
- **Voice Notes:** Mentors and students will heavily utilize voice notes for code reviews to explain *concepts* rather than typing out exact code fixes.
- **Validation:** When a student correctly answers a peer's doubt, mentors will publicly validate them (e.g., "✅ Good catch") to build confidence.

Gamification & Clout

- **The Bounty Board:** Milestones are treated as "bounties." Students must post screenshots of their successful Postman requests or pgAdmin tables to the group.
- **Top Debuggers:** At the end of each week, the most helpful students will be shouted out on a "Top Debuggers" leaderboard and noted for future club leadership considerations.